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Entity-Relationship Model:

COMPSCI 2DB3

Foreword:
This document consists of three parts. First we have an annotated description of the database to highlight all the entities, attributes, relationships, constraints and irrelevant information. Next is a ER diagram modeling the database. And finally, a description of the diagram that highlights the choices I made.

Analysis:

In the analysis, I use the following annotations:

An entity	an entity like object
An attribute	an attribute of some entity like object
A relationship	a relationship between entity like objects
A constraint	a constraint on data
Not relevant	the piece of text is not relevant to the er model

Description:

A local community group is brainstorming with local artists to support artistry and share artwork with the community in a novel way via an art library that enables the local community to learn about the artists, borrow their art, and support their art (as part of the library subscription fees will return to the artists).

The main component of the art library will be the art catalog that maintains information on the art pieces[1] that can be borrowed from the library. In addition, the art catalog information will be used to promote art pieces and artists via the library website.

Central t[1]o the art catalog are art pieces. Each art piece has a unique main artist and title (hence, two distinct artists can both make distinct art pieces named “tomorrow”).[2] Besides the main artist, multiple other artists can contribute to the art piece. In that case, a strict order of the individual contributions needs to be maintained (e.g., main artist, second contributor, third contributor, and so on).[3] Typically, art pieces also have a time and place of creation (for now, the library expects to only hold newly created pieces for

[1] The database is of a singular library that contains a singular art catalog which contains art pieces. Because there is a singular library, and singular catalog, they can be omitted from the model as it would be redundant.

[2] An entity art piece with attributes main artist and title which can both be used as the unique identifiers as 2 art pieces can have the same title, and 2 art pieces can have the same artist.

[3] An entity artist which has a many(total) to many(total) relationship with art pieces. It is a total relation since every art piece has to have an artist, and every artist has to have contributed to some art piece. To account for the strict order of contributors, the relation will have its own attribute “rank”.

which these details are always available). For display purposes, the art library will maintain the physical dimensions (e.g., size of a painting) of each art piece.[4]

For some types of art pieces, other details are maintained. Initially, the library will be working with talented painters, sculptors, and photographers.[5] For both paintings and photo prints, the library will catalog the category (e.g., still life, portrait, landscape, and so on) and the coloring (e.g., natural colors or a black-and-white color tone), and the type of support material (e.g., canvas or wood) on which the painting was made or photo was printed For paintings, the library will in addition catalog the type of paint used (e.g., oil-based or water-based).[6] For sculptures, the library will catalog its weight, whether it can be displayed indoor or outdoor, and the main material used.[7] Finally, each photo print will be related

to details on the original photo the print was based on (e.g., the art piece is a reprint on canvas made in 2022 by Freda, based on a photo made by the photographer Xanthe in 1983).[8] In the catalog, photos (the original negatives or the original digital photo file) are non-physical art pieces: they cannot be borrowed from the library and do not have physical dimensions, but do have titles, an artist that created them, other contributors, and a time and place of creation.[9]

For each artist, the library maintains a profile page that is used to highlight the artist and via which one can find all art pieces of that artist in the catalog.[10] Each artist profile displays their name, their current location (e.g., Hamilton if that is the location of their main atelier), and their age.[11] In addition, the artist can add links to external resources such as personal websites, Instagram pages, YouTube channels, and so on.[12]

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[4] Each art piece will also have the following attributes; time of creation, place of creation, and physical dimensions

[5] Art pieces will have a hierarchical scheme that breaks down into physical, and non physical pieces(entities). The physical pieces consist of entities; paintings, sculptures, and photo prints whereas the non physical pieces will only be the entity photos(digital) There are no primary keys on these as they are all child entities of art pieces.

[6] The photo print and painting entity will have attributes; category, coloring, and support material. The paintings will have an additional paint attribute.

[7] The sculpture entity will have attributes; weight, outdoor flag(where true means the piece can be stored outdoors, and false means it cannot be stored outdoors), and main material used

[8] A one(total) to many(partial) relation from entity photo print to entity photo(digital). It is a total one because each print has to be based on a digital photo, but cannot be based on more than one. And it is a partial many because there can be many prints based on the original, but there can exist some digital photos that do not have any prints.

[9] Photos(digital) are entities that “Is a” non physical art piece. They do not associate with the borrowed and/or reserved relations, and do not have any dimension attributes. Therefore the dimension attributes are only associated with physical pieces. Since they are child entities of art pieces, they still reserve the title, artist, place, and time attributes.

[10] The entity artist owns a new entity profile page which is why profile page is a weak entity. Artist also has a one(total) relationship with profile pages since “for each” implies every artist has a profile page. And it is assumed that no artist will have more than 1 profile page.

[11] A profile page has attributes; name (which can be a composite attribute that contains a first, middle, and last name), current location, age (which can be derived from the artists date of birth), as well as a profile page id which will be used as the partial key.

[12] Since an artist can in addition add extra links to other sites, a many(partial) to many(total) relation is made between profile pages and a links entity. Many profile pages can contain multiple links (or none) and are therefore many(partial). And it's possible that the same link can relate to multiple different artist pages, however there isn't a case where a link does not relate to an artist's profile so therefore many(total).

Art pieces can be grouped together, e.g., they can be part of the same collection(s), or they can be part of a group of art pieces made within a collaboration.[13] Each such a group has a title, type, and description. Several groups can have the same title, types, and/or descriptions.[14]

Finally, the library will have members that can borrow art pieces.[15] The library plans to a pre-existing system to manage memberships and payment details.[16] This system will assign a unique member id to each member.[17] That system will not manage the borrowed and reserved art pieces, however.[18] Each borrowed work is associated with an art piece, the member that borrowed it, and the time period during which the work is borrowed.[19] Each reserved work is associated with an art piece, the member that wants to borrow it, and the time when the reservation was placed[20] (reserving art pieces works on a first-come first-serve basis).[21] Members can borrow the same art piece multiple times and members can indefinitely renew their borrow period unless other members have reserved the piece[22] (each renew is re gistered as a separate borrow).[23]

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[13] A new entity collection s is introduced. A many(partial) to many(total) relation between art pieces and collection of art pieces is made. Many pieces can be grouped together, but possible for an art piece to not be a part of a collection (partial). If a collection is present, it has to have an art piece (total).

[14] The collection entity has attributes title, type, and description. However, none of the above can be the primary key and so therefore a collection id is introduced.

[15] A new entity member is introduced that will have a relation with art pieces.

[16] Irrelevant information that adds nothing to the model.

[17] Members will have a primary key attribute: member id.

[18] Irrelevant information

[19] The borrow relationship between member and art piece is many(partial) to many(partial) because many members can borrow art, and many art can be borrowed by a member. They are both partial because some members may not borrow any art, and some art may never be borrowed. Additionally, the borrowed relationship will contain the time period it was borrowed, which can be derived through two attributes: start time and end time.

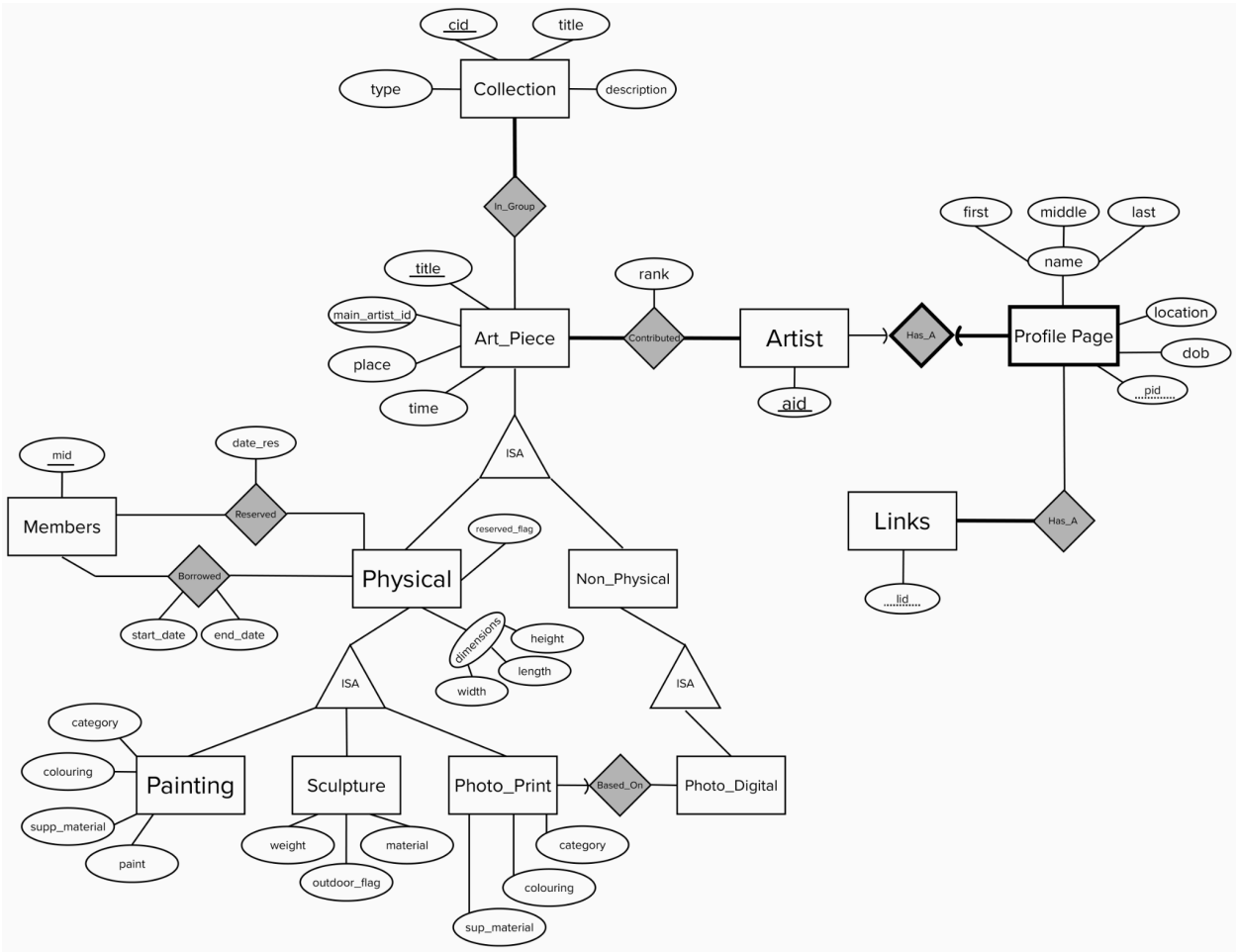
[20] Similar to the borrow relationship, a new relationship reserved is introduced. Reserved is also many(partial) to many(partial) for the same reasons. However the relation has only one attribute: time of the reservation which will indicate when the art piece is being reserved.

[21] A constraint that needs a reserved flag on all reservable items. Therefore the physical art piece entities will contain a reserve flag. If it's false, then the first member to reserve it sets it to true. Then returns to false once the member returns it.

[22] A constraint in which someone cannot borrow a piece if it is reserved for the time the member wishes to borrow.

[23] Irrelevant information.

Diagram:



Model Solution:

The diagram represents a database to illustrate an art catalog in a library. Since this is a model for a unique library, it is not extendable to other libraries and therefore does not need to contain a library entity. Furthermore, there is only one catalog for the library, hence there is also no need for a catalog entity. We begin with art pieces which are represented by an entity **Art_Piece**. Since there can be multiple art pieces with the same artist, and multiple art pieces with the same title, the entity **Art_Piece** has two unique identifiers: *title*(of type text) and *main_artist_id*(of type text). **Art_Piece** also has attributes *place* (of type text) and *time* (of type timestamp). Since all **Art_Piece**'s can be grouped into different collections, we have a relation between **Art_Piece** and entity **Collection** via a relation In_group which is many(partial) to many(total). Every collection entity must have an art piece it is related to, however an art piece does not have to be related to a collection.

Collection has multiple attributes including a uniquely randomly generated primary key *cid* (of type integer), as well as *title*(of type text)*, description*(of type text)*,* and *type*(of type text).

To account for all artists that contribute to the library, we include an entity **Artist** which has a many(total) to many(total) relation Contributed with **Art_Piece.** Since all art pieces must have a strict order (1st, 2nd, 3rd) of contributors to an art piece, we include an attribute to the relationship *rank*(of unique type integer). The key constraint here is that the main artist will always have a rank of 1. It is a many to many total relation since every art piece needs to have an artist, but can have multiple. And if you are an artist in the library, then that means you must have contributed to an art piece (total), and artists can contribute to multiple pieces.**Artist** will also have 1 unique identifier *aid*(of type unique integer).

Each artist also owns a single profile page which is represented by a weak entity **Profile_Page.** Because of the 1 profile page per artist constraint, the relation Has_A from **Artist** to **Profile_Page** is one(total) to one(total). **Profile_Page** has a partial key *pid*(of type unique integer). This entity also has a composite attribute *name* which contains attributes *first*(of type text)*, middle*(of type text)*, last*(of type text). It also has a *location*(of type text) attribute, and to account for the age of each artist, we can include a derived attribute *dob* or “date of birth”(of type date) which can easily calculate the age of the artist. Finally, each profile page can contain external links to other websites that the artist wishes to promote. Since an artist can include multiple links, I’ve decided to make it a separate entity.

The entity **Links** shares a many(total) to many(partial) relation Has_A with **Profile_Page**. Every link must be related to a profile page, and can be related to multiple. Every profile page can have 0+ links on their page. **Links** has 1 partial identifier *lid*(of type unique integer)

Since art pieces can either be physical, or non physical, I incorporate a hierarchical ISA relation that divides **Art_Piece** into 2 sub entities **Physical** and **Non_Physical.** Neither entity needs a unique id since they are child entities of **Art_Piece** and therefore absorb all **Art_piece** attributes. This relation is done to allow for the constraint that says only physical art pieces can be borrowed / reserved. The **Physical** art pieces have attributes *reserved_flag*(of type boolean) which is set to true if a member has reserved the item and false if it has not yet been reserved. It also has a composite attribute *dimension* which has *height*(of type integer)*, width*(of type integer)*,* and *length*(of type integer)*.*

Members of the library are represented by the entity **Members** which has a single unique identifier *mid* (of type unique integer). **Members** have a many(partial) to many(partial) relation Reserved with **Physical** entities. The Reserved relation has an attribute *date_res*(of type date) which is used to keep track of the date of the reservation. **Members** and **Physical** also share another many(partial) to many(partial) relation Borrowed which also has two attributes *start_date*(of type date) and *end_date*(of type date) which keep track of the start and end of the borrow period. Furthermore, the constraint of multiple borrows being allowed is irrelevant for this model, however the constraint of reservations having priority over borrows is implemented via the *reserved_flag* attribute that was implemented to **Physical**.

Furthermore, **Physical** entities can also be further broken down into 3 sub entities **Painting, Sculptures,** and **Photo_Prints** via another ISA relation. For a cleaner model, I also added an ISA relation for **Non_Physical** entities which is just broken down into a single entity **Photo_Digital.**

Paintings do not have a primary identifier as it is a child entity of **Art_Piece** however it does have 4 attributes: *category*(of type text)*, coloring*(of type text)*, support_material*(of type text)*,* and *paint*(of type text)*.*

Sculptures is also a child of **Art_Piece** and hence does not have a unique identifier. However it does have attributes: *weight*(of type double)*, material*(of type text)*, and outdoor_flag*(of type boolean) to account for the constraint of whether or not a sculpture can be displayed indoor and or outdoor.

Since the library contains both **Photo_Prints** and **Phot_Digital,** I have separated the two into the respective physical and non -physical subclasses. **Photo_Prints** are borrowable and therefore physical. **Photo_Prints** are a child of **Art_Pieces** and has attributes: *category*(of type text)*, coloring*(of type text)*, support_material*(of type text). **Photo_Digital** are not borrowable and therefore is a subclass of non-physical entities. **Photo_Digital** does not have any attributes aside from the ones inherited from **Art_Pieces**. To account for the constraint that every **Photo_Print** has to be based on a **Photo_Digital** I incorporated a one(total) to many(partial) relation Based_On. It is one(total) since every **Photo_Print** has to be based on exactly 1 **Photo_Digital**, and many(partial) since every **Photo_Digital** can have 0+ **Photo_Prints**.

The following constraints are not expressed in the diagram:

- The attribute *mid* of entity **Members** is unique
- In relation Borrowed, values for start_time has to be before values for end_time
- In relation Contributed, a strict order is maintained
- In entity **Physical**, the attribute *reserved_flag* works on a first come first serve basis
- The relation Borrowed can be indefinitely renewed by the same member unless the piece is already reserved.

I have also attached the diagram as a separate pdf.